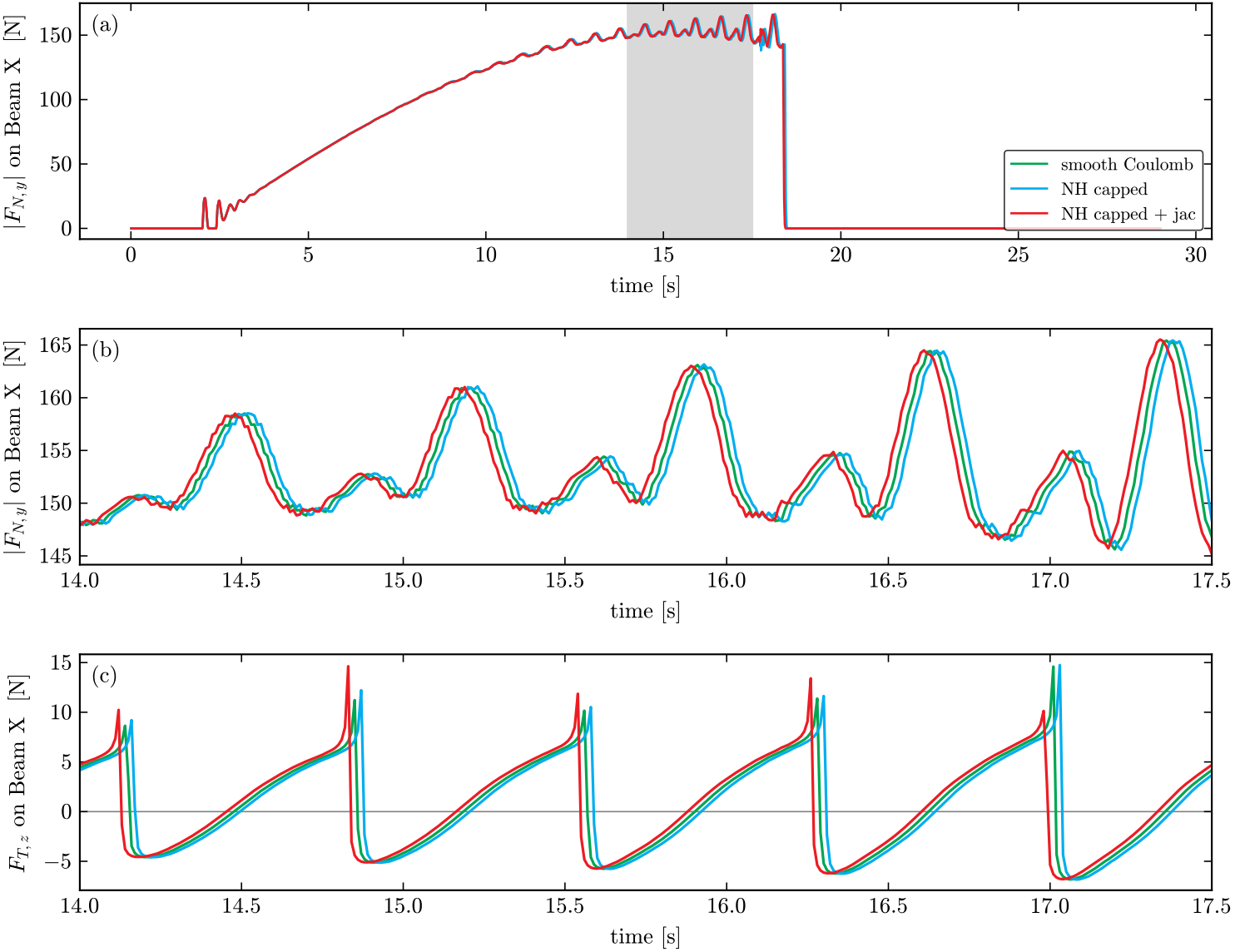
(a)
$|F_{N,y}|$ on Beam X [N]
150
100
50
0
0
5
10
15
20
25
30
time [s]
smooth Coulomb
NH capped
NH capped + jac
(b)
$|F_{N,y}|$ on Beam X [N]
165
160
155
150
145
14.0
14.5
15.0
15.5
16.0
16.5
17.0
17.5
time [s]
(c)
$F_{T,z}$ on Beam X [N]
15
10
5
0
−5
14.0
14.5
15.0
15.5
16.0
16.5
17.0
17.5
time [s]